BUDDHA SERIES

(Unit Wise Solved Question & Answers)

Course – BBA 2nd Sem

College – Buddha Institute of Management

(DDU Code-1212)

Department: Business Administration
Subject: Basics of Statistics
Faculty Name: Mr. Abhishek Kumar Gupta

Unit – I

1.What is the statistics?

Ans. Statistics is a branch of applied mathematics that involves the collection, description, analysis, and inference of conclusions from quantitative data. The mathematical theories behind statistics rely heavily on differential and integral calculus, linear algebra, and probability theory.

Those who work with statistics are referred to as statisticians. They're particularly concerned with determining how to draw reliable conclusions about large groups and general events from the behavior and other observable characteristics of small samples. These samples represent a portion of the large group or a limited number of instances of a general phenomenon.

Statistics are used in virtually all scientific disciplines such as the physical and social sciences as well as in business, medicine, the humanities, government, and manufacturing. Statistics is a branch of applied mathematics including calculus and linear algebra that developed from the application of mathematical tools to probability theory.

2.write the types of statistics.

Ans. The two major areas of statistics are descriptive statistics and inferential statistics.

(i) Descriptive statistics: It describes the properties of sample and population data.Descriptive statistics focus mostly on the central tendency, variability, and distribution of sample data. Central tendency refers to the estimate of the characteristics, a typical element of a sample or population. It includes descriptive statistics such as mean, median, and mode.

Variability refers to a set of statistics that show how much difference there is among the elements of a sample or population along the characteristics measured. It includes metrics such as range, variance, and standard deviation.The distribution refers to the overall "shape" of the data.

This can be depicted on a chart such as a histogram or a dot plot and includes properties such as the probability distribution function, skewness, and kurtosis.Descriptive statistics can also describe differences between observed characteristics of the elements of a data set. They can help us understand the collective properties of the elements of a data sample and form the basis for testing hypotheses and making predictions using inferential statistics.

There are four main types of descriptive statistics: measures of frequency, central tendency, variability or dispersion, and measures of position.

The most commonly reported descriptive statistics are the mean and range.

(ii) Inferential statistics: It uses the properties of sample & population to test hypotheses and draw conclusions. **ypes of Inferential Statistics**

There are different types of inferential statistics, including:

Confidence Interval

It is a range of values that is likely to contain the true value of the population parameter with a certain level of confidence. Mathematically it is defined as the mean of your estimate plus and minus the variation in that estimate. It is useful as it provides a range of plausible values for a population parameter based on the sample data.

Regression Analysis

It is a technique for analyzing the relationship between two or more variables. In simple terms, it predicts the value of one variable based on the other variables.The regression analysis aims to find the best-fit line that can explain the relationship between variables. There are two types of regression analysis: Simple Regression and Multiple Regression.

Hypothesis Testing

A statistical technique used to evaluate (or verify) the hypothesis (or assumptions) of the sample dataset about the population dataset. Hypothesis Testing involves:

- formulation of Null Hypothesis
- formulation of Alternate Hypothesis
- select the sample from the population
- perform the statistical test
- check the p-value
- if the p-value is less than a significant level, then reject the null hypothesis
- if the p-value exceeds the significance level, we fail to reject the null hypothesis.

Analysis of Variance

Analysis of Variance (ANOVA) is a statistical method to analyze the difference between two or more groups. It is a type of hypothesis testing that compares the variance of different groups to determine if the mean difference between them is statistically significant.

There are two types of ANOVA: One-way ANOVA (includes only one variable) and Two-way ANOVA (used when there are two independent variables).

3. What is importance & scope of statistics?

Answer. Statistics plays a crucial role in business and economic decision making. Here are some key reasons why statistics is important in these fields:

1. **Data Analysis:** Statistics helps in analyzing large amounts of data to identify patterns, trends, and relationships. This analysis provides valuable insights into market conditions, consumer behavior, and economic indicators, which are essential for making informed business decisions.
2. **Forecasting and Planning:** By using statistical techniques such as time series analysis and regression analysis, businesses can forecast future trends and plan accordingly. This helps in setting realistic goals, estimating demand, optimizing resources, and making strategic decisions.
3. **Risk Assessment and Management:** Statistics enables businesses to assess and manage risks effectively. Through techniques like probability distributions and hypothesis testing, businesses can evaluate the likelihood of different outcomes and make decisions that minimize risks and maximize returns.
4. **Market Research:** Statistics is vital in market research, where data is collected and analyzed to understand consumer preferences, market trends, and competitor analysis. This information helps businesses develop effective marketing strategies, launch new products, and gain a competitive edge.
5. **Performance Evaluation:** Statistics provides tools for measuring and evaluating business performance. Key performance indicators (KPIs) are used to track progress, identify areas for improvement, and make data-driven decisions to enhance efficiency and profitability.

6. **Economic Policy Making:** Statistics is essential for economic policy making at both macro and micro levels. Governments and policymakers rely on statistical data to assess economic conditions, monitor inflation, unemployment rates, and GDP growth, and formulate policies that promote economic stability and growth.
7. **Decision Making Under Uncertainty:** In business and economics, decisions are often made in uncertain environments. Statistics provides techniques like probability theory and decision analysis to make rational decisions by considering the probabilities and potential outcomes.

Finance

The financial analysts are able to guide their clients about stocks investments and give their recommendations by using a variety of statistical information and tools. A lot of financial data is reviewed by analysts like price/earnings ratio and dividend yields. Analysts make comparisons between an individual stock and the average of stock market and then come to analyze if a stock is over/underpriced and should be bought or not.

Marketing

Statistics have a lot of scope in marketing research applications. There are organizations like KPMG, NIELSEN etc which specialize in providing national-international level surveys based on marketing. These collect data from various point of sale of super markets like Big Bazaar, Spencers, EasyDay, Vishal Mega Mart etc and analyze the data contained in the invoices generated by terminals using various statistical tools. This type of data is generally not available freely thus such organizations have to spend huge amount to get this data. Similarly, special pricing days, offers, cash backs etc data from all other type of promotional activities is studied and analyzed using statistical tools to come to conclusions to be presented in Survey reports. A careful study of these reports may prove to be quite helpful in formulating marketing strategies for the future.

Economics

How the economy will grow? Whether it will grow or it will fall back? These are some of the forecasts the economists need to make. With statistics, making such forecasting is possible by analyzing the economical factors using Statistics as a lot of statistical information is used by economists in making such forecasts. For instance, while doing trading and buying/selling a particular commodity, the person likes to see how a particular commodity is doing over a past period of time and only then they will be able to decide whether to sell it or retain it. Statistics help in making such decisions. Moreover, statistics is also used to find the relationship between supply and demand, the inflation rate, imports and exports, the per capita income etc.

Computer Science

In computer science, statistics is used for a number of things, including data mining, data compression and speech recognition. Other areas where statistics are use in computer science include vision and image analysis, artificial intelligence and network and traffic modeling. Data mining is performed with the help of statistics by using functions to find irregularities or inconsistencies within data. Data compression uses statistical algorithms to compress data. Statistics are also used in network traffic modeling, whereby available bandwidth is exploited to be usable while the use of statistical programs avoids network congestion. Artificial intelligence tries to simulate human thought using algorithms that are similar to voice recognition or translation software. Other statistical uses in computer science include quality management, software engineering, storage and retrieval processes and software and hardware engineering and manufacturing. Algorithms have become necessary in many facets of computer programming and data mining.

Education

Education is a field where evaluation of students, teachers and other entities is a continuous activity. The collection, organization, analysis, interpretation and presentation of data are the activities that need to be conducted and this data is always huge and thus requires the use of statistical tools. For instance, calculating the average marks of a class can be represented using bar graph and histograms, cumulative frequency distribution curve, pie charts etc. Similarly, the report prepared after the processing of results is supposed to present the data statistically in graphical format. There also, statistical methods come very handy.

In Daily Life

Statistics is used in every walk of life like weather forecasting, detecting an illness, medical research, making policies for business sales, deciding on offers and discounts, in political campaigns, to keep track of sales, and promotion of insurance policies, analyzing the stock market and what not. Thus, statistics find its utility in every industry we can think of and its methods are a great help in analyzing different business situations and data analysis.

4. what is the limitations of statistics?

Answer. **Limitations of Statistics**

The statistic is restricted by certain limitations:

1. **Useful for numerical studies only:**

Statistics can work upon only numeric data. By numeric data, we mean data that is available in numbers that can be measured quantitatively and expressed numerically. It cannot be applied to all types of data.

2. **Doesn't consider individual but population:**

The results presented by statistical analysis are based on a specific group as it is not feasible to take into account everything, thus that specific group which is called population is considered on behalf for the whole group but the results produced by the statistical methods are said to be applied on the whole group i.e they are not restricted to the population only. For example, in the statement 03 in every 10 Indian adults are obese, it shows the results which are found by considering all individuals.

3. **Dependent on estimates and rough calculations:**

The results produced by statistical methods are based on a specific population and not for every member of the universe and thus in certain situations are not reliable.

4. **May promote false conclusions by deliberate manipulation of figures and unscientific handling:**

Results generated from statistical methods are numerical in nature and depicted in figures that can be modified. The data can be represented graphically which is out of the context.

5. **More than one method for a single problem**:

In statistics, there are several methods available that an analyst may use to solve a single problem and results found through each of the methods may have different results. For example, Variation can be found by quartile deviation, mean deviation or standard deviations and results vary in each case.

6. **Provide a basis of judgement but not the whole judgement**:

As stated earlier, statistical methods work only on numerical data therefore it may happen that certain factors which are very important but cannot be measured/converted in numerical terms remain ignored and away from the analysis.

7. **Misleading results:**

Statistics may give misleading results if proper attention is not paid while data collection, analysis and interpretation.

8. E**xpert knowledge required:**

To handle statistics, one must have the required expertise and command on the subject and must be able to apply their wisdom while selecting statistical methods.

5. What is the statistical units?
Answer. A statistical unit is a unit of observation or measurement for which data are collected or derived. The statistical unit is therefore the basic element for compiling and tabulating statistical data.
A statistical unit is an object of statistical survey and the bearer of statistical characteristics. The statistical unit is the basic unit of statistical observation within a statistical survey.

6. What is Primary Data?

Ans: Primary data is a type of data that is collected by researchers directly from main sources through interviews, surveys, experiments, etc. Primary data are usually collected from the source—where the data originally originates from and are regarded as the best kind of data in research.The sources of primary data are usually chosen and tailored specifically to meet the demands or requirements of particular research. Also, before choosing a data collection source, things like the aim of the research and target population need to be identified.

For example, when doing a market survey, the goal of the survey and the sample population need to be identified first. This is what will determine what data collection source will be most suitable—an offline survey will be more suitable for a population living in remote areas without an internet connection compared to online surveys.

7. What is the methods of collecting primary data?

Ans. Primary Data Collection Methods: Primary data collection methods are different ways in which primary data can be collected. It explains the tools used in collecting primary data.

Primary data or raw data is a type of information that is obtained directly from the first-hand source through experiments, surveys or observations. The primary data collection method is further classified into two types. They are

(i) Quantitative Data Collection Methods

(ii)Qualitative Data Collection Methods

Let us discuss the different methods performed to collect the data under these two data collection methods.

(i)Quantitative Data Collection Methods

It is based on mathematical calculations using various formats like close-ended questions, correlation and regression methods, mean, median or mode measures. This method is cheaper than qualitative data collection methods and it can be applied in a short duration of time.

(ii)Qualitative Data Collection Methods

It does not involve any mathematical calculations. This method is closely associated with elements that are not quantifiable. This qualitative data collection method includes interviews, questionnaires, observations, case studies, etc. There are several methods to collect this type of data. They are

Observation Method

Observation method is used when the study relates to behavioural science. This method is planned systematically. It is subject to many controls and checks. The different types of observations are:

Structured and unstructured observation

Controlled and uncontrolled observation

Participant, non-participant and disguised observation

Interview Method

The method of collecting data in terms of verbal responses. It is achieved in two ways, such as

Personal Interview – In this method, a person known as an interviewer is required to ask questions face to face to the other person. The personal interview can be structured or unstructured, direct investigation, focused conversation, etc.

Telephonic Interview – In this method, an interviewer obtains information by contacting people on the telephone to ask the questions or views, verbally.

Questionnaire Method: this method, the set of questions are mailed to the respondent. They should read, reply and subsequently return the questionnaire. The questions are printed in the definite order on the form. A good survey should have the following features:

Short and simple

Should follow a logical sequence

Provide adequate space for answers

Avoid technical terms

Should have good physical appearance such as colour, quality of the paper to attract the attention of the respondent

Schedules

This method is similar to the questionnaire method with a slight difference. The enumerations are specially appointed for the purpose of filling the schedules. It explains the aims and objects of the investigation and may remove misunderstandings, if any have come up. Enumerators should be trained to perform their job with hard work and patience.

8. Write the methods of collecting secondary data.

Answer: Secondary data is data collected by someone other than the actual user. It means that the information is already available, and someone analyses it. The secondary data includes magazines, newspapers, books, journals, etc. It may be either published data or unpublished data.

(i) Published data are available in various resources including

Government publications, Public records, Historical and statistical documents, Business documents, Technical and trade journals

(ii) Unpublished data includes

Diaries, Letters, Unpublished biographies, etc.

9. What is the graphical representations of data?

Answer: Graphical Representation is a way of analysing numerical data. It exhibits the relation between data, ideas, information and concepts in a diagram. It is easy to understand and it is one of the most important learning strategies. It always depends on the type of information in a particular domain. There are different types of graphical representation. Some of them are as follows:

Line Graphs – Line graph or the linear graph is used to display the continuous data and it is useful for predicting future events over time.

Bar Graphs – Bar Graph is used to display the category of data and it compares the data using solid bars to represent the quantities.

Histograms – The graph that uses bars to represent the frequency of numerical data that are organised into intervals. Since all the intervals are equal and continuous, all the bars have the same width.

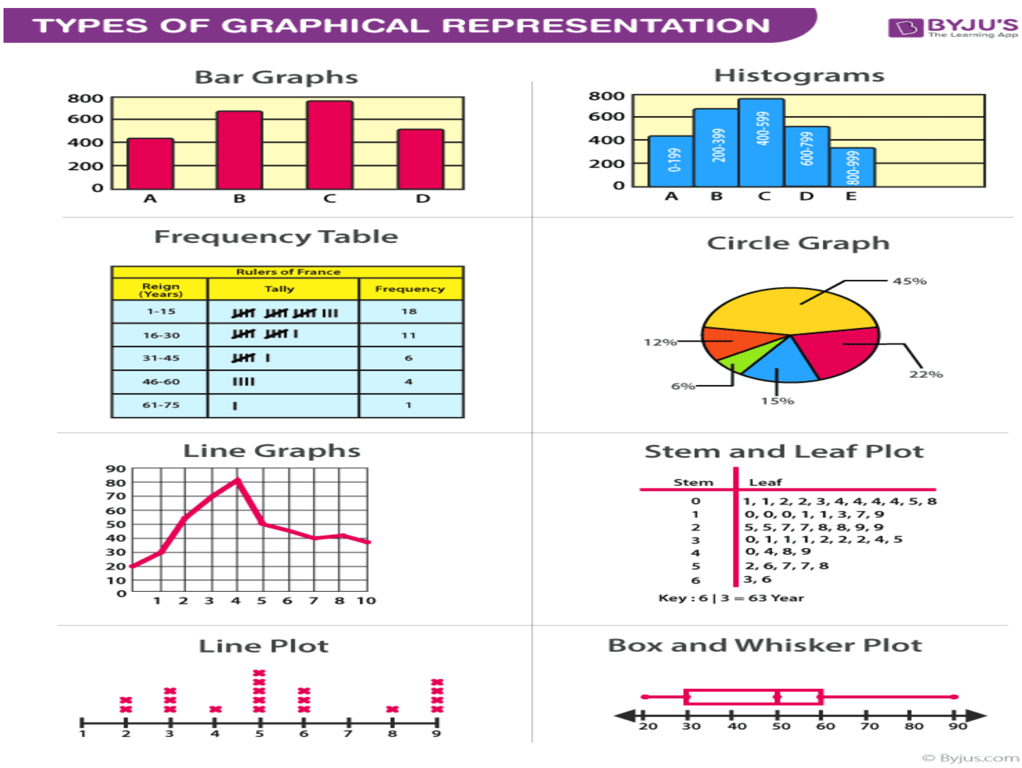
Line Plot – It shows the frequency of data on a given number line. ‘ x ‘ is placed above a number line each time when that data occurs again.

Frequency Table – The table shows the number of pieces of data that falls within the given interval.

Circle Graph – Also known as the pie chart that shows the relationships of the parts of the whole. The circle is considered with 100% and the categories occupied is represented with that specific ercentage like 15%, 56%, etc.

Stem and Leaf Plot – In the stem and leaf plot, the data are organised from least value to the greatest value. The digits of the least place values from the leaves and the next place value digit forms the stems.

Box and Whisker Plot – The plot diagram summarises the data by dividing into four parts. Box and whisker show the range (spread) and the middle (median) of the data.



10. Explain frequency distribution & its type.

Answer: A frequency distribution is a way to organize data and see how often each value appears. It shows how many times each value or range of values occurs in a dataset. This helps us understand patterns, like which values are common and which are rare. Frequency distributions are often shown in tables or graphs, and make it easier to analyze and conclude data.

Types of frequency distribution

(i) Individual Frequency Distribution

That frequency distribution in which each item or value is shown separately, is known as individual frequency distribution. Example: the marks obtained by 7 students in tutorial are 3,10,4,9,0,1,9.

(ii) Ungrouped Frequency Distribution:

In Ungrouped Frequency Distribution, all distinct observations are mentioned and counted individually. This Frequency Distribution is often used when the given dataset is small.

Example: Make the Frequency Distribution Table for the ungrouped data given as follows: 10, 20, 15, 25, 30, 10, 15, 10, 25, 20, 15, 10, 30, 25

Solution: As unique observations in the given data are only 10, 15, 20, 25, and 30 with each having a different frequency. Thus, the Frequency Distribution Table of the given data is as follows:

Value	10	15	20	25	30
frequency	4	3	2	3	2

(iii) Grouped Frequency Distribution

In Grouped Frequency Distribution observations are divided between different intervals known as class intervals and then their frequencies are counted for each class interval. This Frequency Distribution is used mostly when the data set is very large.

Example: Make the Frequency Distribution Table for the ungrouped data given as follows:

23, 27, 21, 14, 43, 37, 38, 41, 55, 11, 35, 15, 21, 24, 57, 35, 29, 10, 39, 42, 27, 17, 45, 52, 31, 36, 39, 38, 43, 46, 32, 37, 25

Solution:

As there are observations in between 10 and 57, we can choose class intervals as 10-20, 20-30, 30-40, 40-50, and 50-60. In these class intervals, all the observations are covered and for each interval, there are different frequency which we can count for each interval.

Thus, the Frequency Distribution Table for the given data is as follows:

C.I.	10-20	20-30	30-40	40-50	50-60
F	5	8	12	6	3

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Subject: Basics of Statistics

Faculty Name: Mr. Abhishek Kumar Gupta

Unit – II

Question1.The central tendency is stated as the statistical measure that represents the single value of the entire distribution or a dataset. It aims to provide an accurate description of the entire data in the distribution.

Class Marks	10-25	25-40	40-55	55-70	70-85	85-100
Number of Students	2	3	7	6	6	6

Solution: Given-

Class Interval	10-25	25-40	40-55	55-70	70-85	85-100
Number of Students	2	3	7	6	6	6

for each class interval, we need to find the midpoint (class mark) that serves as the representative of the whole class.

For example, for the first class interval, 10-25, the class mark is:

Class Mark = (Upper class limit + lower class limit)/2

Class Mark = (25+10)/2 = 17.5

Similarly, find the class mark for all the intervals.

Therefore, the mean of the marks obtained by the students is given as:

Class Interval	Number of students (fi)	Class Mark (xi)	fixi
10-25	2	17.5	35
25-40	3	32.5	97.5
40-55	7	47.5	332.5
55-70	6	62.5	375
70-85	6	77.5	465
85-100	6	92.5	555
Total	Σfi = 30		Σfixi = 1860

Therefore, Mean, $\bar{x}$ = 1860/30 = 62

The mean value obtained using the direct method is 62.

The points scored by a Kabaddi team in a series of matches are as follows:

17, 2, 7, 27, 15, 5, 14, 8, 10, 24, 48, 10, 8, 7, 18, 28

Question 2. Find the median of the points scored by the team.

Solution: Given, 17, 2, 7, 27, 15, 5, 14, 8, 10, 24, 48, 10, 8, 7, 18, 28

Arranging the points scored by the team in ascending order, we get;

2, 5, 7, 7, 8, 8, 10, 10, 14, 15, 17, 18, 24, 27, 28, 48

Number of observations = n = 16

Median = M = (N+1) / 2

= (17/2)th value

= 8.5th value

= (8^{th} value + 9^{th} value) / 2

= (10+14) / 2

= 24 / 2

= 12

Question 3. Calculate the median for the following distribution.

Class interval	Frequency
40 – 44	1
45 – 49	5
50 – 54	9
55 – 59	12
60 – 64	7
65 – 69	2

Solution:

The given frequency distribution has discontinuous classes so let us convert them as continuous classes and calculate the cumulative frequency.

Class intervals	Frequency	Cumulative frequency
39.5 – 44.5	1	1
44.5 – 49.5	5	6
49.5 – 54.5	9	15
54.5 – 59.5	12	27
59.5 – 64.5	7	34
64.5 – 69.5	2	36

N = 36

N/2 = 36/2 = 18

Cumulative frequency greater than and nearer to 18 is 27 which belongs to the class interval 54.5 – 59.5.

Thus, median class = 54.5 – 59.5

L_1 = Lower limit of the median class = 54.5

N = Sum of frequencies = 36

c.f. = Cumulative frequency of the class preceding the median class = 15

f = Frequency of median class = 12

h = Class height = 5

Median = + [(N/2 – cf)/f] × h

= 54.5 + [(18 – 15)/12] × 5

= 54.5 + [(3 × 5)/12]

= 54.4 + (5/4)

= 54.5 + 1.25

= 55.75

Therefore, the median of the given distribution is 55.75.

Question 4: For any given data, the mean is 45.5, and the median is 43. Find the modal value.

Solution: We know that,

Mode = 3 Median – 2 Mean

∴ Mode = 3 × 43 – 2 × 45.5

= 129 – 91 = 38.

Mode = 38.

Question 5. Find the mode of the following data distribution

Classes	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	12	14	10	13	14	18	10

Solution: Since 18 is the maximum frequency of given data**.**

So, the modal class is 60-70.

l = 60

f_k = 18

f_{k-1}= 14

f_{k+1}= 10

h = 70 – 60 = 10

Mode = $L_1 + [(f_1 – f_0) / (2f_1 – f_0 – f_2)] h$

= 60 + [(18 – 14)/(2 × 18 – 14 – 10)] × 10

= 60 + [4/12] × 10

= 60 + 10/3

= 63.333 (approx).

Question 6: Find the mode of the following data distribution:

Class-Interval	10-20	20-30	30-40	40-50
Frequency	12	35	45	25

Solution: Given –

Class-Interval	10-20	20-30	30-40	40-50
Frequency	12	35	45	25

Since 45 is the maximum frequency.

So, the modal class is 30-40, then

L_1 = 30

f_k = 45

f_{k-1}= 35

f_{k+1}= 25

h = 40 – 30 = 10

Mode= $L_1 + [(f_1 – f_0) / (2f_1 – f_0 – f_2)] h$

= 30 + [(45 – 35)/(2 × 45 – 35 – 25)] × 10

= 30 + [10/30] × 10

= 30 + 10/3

= 100/3 = 33.333.

Question7. The marks scored by the students of class 10 are 45, 39, 55, 63, 49, 92, and 79. Find the range of the given dataset**.**

Solution:

Given dataset values: 45, 39, 55, 63, 49, 92, 79.

Now, arrange the given dataset in ascending order.

I.e., 39, 45, 49, 55, 63, 79, 92.

Here,

Highest data value = 92

Lowest data value = 39.

Therefore, Range = Highest data value – Lowest data value

Range = 92 – 39

Range = 53.

Therefore, the range of the given data is 53

Question 8. Determine the highest value in the data set, if the range equals 40 and the lowest value should be equal to 6.

Solution: Given that the lowest value in the dataset is 6.

And, range = 40.

As we know, Range = Highest data value – Lowest data value.

Now, substitute the values in the formula, and we get

40 = Highest value – 6

Therefore, the highest value = 40 + 6

Highest value = 46.

Question 9.calculate S.D. for following data.

X	5	7	9	11
F	3	5	2	4

Solution: First of all we calculate mean of given data.

x	f	f.x
5	3	15
7	5	35
9	2	18
11	4	44
	14	Σf.x = 112

Mean = $\bar{x} = (\Sigma f.x) / \Sigma f$

$= 112 / 14$

$= 8$

Now, we calculate standard deviation.

X	f	$(x - \bar{x})$	$(x - \bar{x})^2$	$f.(x - \bar{x})^2$
5	3	-3	9	27
7	5	-1	1	5
9	2	1	1	2
11	4	3	9	36
	$\Sigma f = 14$			$\Sigma f.(x - \bar{x})^2 = 70$

$\sigma = \sqrt{[\Sigma f.(x - \bar{x})^2 / \Sigma f]}$

$= \sqrt{70 / 14}$

$= \sqrt{5}$

$= 2.23$

Question10. Determine Karl Pearson's coefficient of skewness for following data.

X	5	7	9	11
F	3	5	2	4

Solution: First of all, we calculate mean, mode & standard deviation of given data.

For mean-

x	f	f.x
5	3	15
7	5	35
9	2	18
11	4	44
	14	$\Sigma f.x = 112$

Mean = $\bar{x} = (\Sigma f.x) / \Sigma f$

$= 112 / 14$

$= 8$

For mode-

By inspection, it is clear that 5 is the maximum frequency of given data.

So, Mode (Z) = 7

For S.D.-

X	f	$(x - \bar{x})$	$(x - \bar{x})^2$	$f.(x - \bar{x})^2$
5	3	-3	9	27
7	5	-1	1	5
9	2	1	1	2
11	4	3	9	36
	$\Sigma f = 14$			$\Sigma f.(x - \bar{x})^2 = 70$

$\sigma = \sqrt{[\Sigma f.(x - \bar{x})^2 / \Sigma f]}$

$= \sqrt{70 / 14}$

$= \sqrt{5}$

$= 2.23$

Skewness = mean – mode

= 8 – 7

= 1

Coefficient of skewness = (mean – mode) / S.D.

= 1 / 2.23

= 0.4484

= 44.84 %

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Unit – III

Question 1. What is the correlation?

Answer: In statistics, correlation describes a relationship between two or more variables, indicating how much they vary together. It's a measure of the degree to which two variables change in relation to each other, expressed as a number called the correlation coefficient.

Question 2. Explain all types of correlation.

Answer: (A) on the basis of direction:

1. Positive Correlation:

When two variables move in the same direction; i.e., when one increases the other also increases and vice-versa, then such a relation is called a **Positive Correlation.** ***For example,*** Relationship between the price and supply, income and expenditure, height and weight, etc.

Simultaneous increase in the value of both the Variables		Simultaneous decrease in the value of both the Variables	
X	Y	X	Y
20	120	60	500
30	170	50	400
40	220	40	300
50	270	30	200
60	320	20	100

2. Negative Correlation:

When two variables move in opposite directions; i.e., when one increases the other decreases, and vice-versa, then such a relation is called a **Negative Correlation.** ***For example,*** the relationship between the price and demand, temperature and sale of woollen garments, etc.

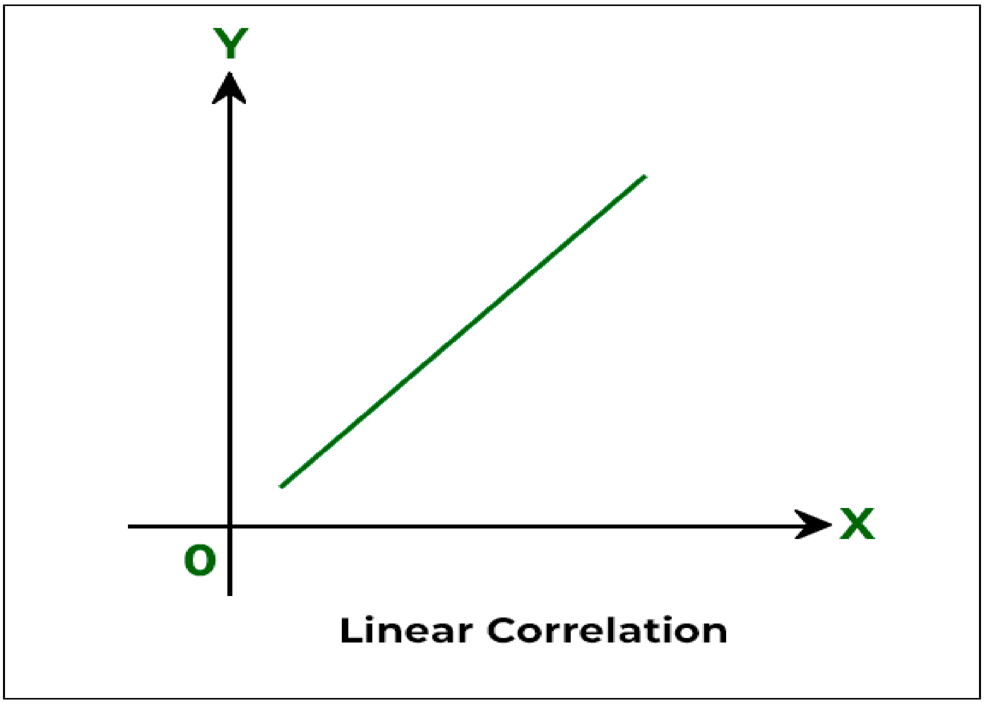
Rise in the value of one variable (X) and fall in other (Y)		Rise in the value of one variable (Y) and fall in other (X)	
X	Y	X	Y
10	100	250	100
20	90	200	k200
30	80	150	300
40	70	100	400
50	60	50	500

(A) on the basis of ratio of change :

. Linear Correlation:

When there is a constant change in the amount of one variable due to a change in another variable, it is known as **Linear Correlation.** This term is used when two variables change in the same ratio. If two variables that change in a fixed proportion are displayed on graph paper, a straight- line will be used to represent the relationship between them. As a result, it suggests a linear relationship.

X	10	15	20	25	30
Y	10	20	30	40	50

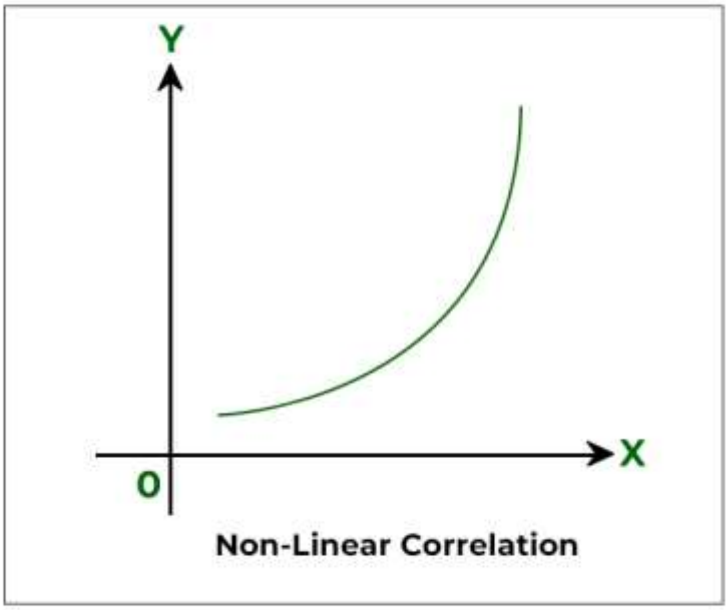


In the above graph, for every change in the variable X by 5 units there is a change of 10 units in variable Y. The ratio of change of variables X and Y in the above schedule is 1:2 and it remains the same, thus there is a linear relationship between the variables.

2. Non-Linear (Curvilinear) Correlation:

When there is no constant change in the amount of one variable due to a change in another variable, it is known as a **Non-Linear Correlation.** This term is used when two variables do not change in the same ratio. This shows that it does not form a straight-line relationship. ***For example*****,** the production of grains would not necessarily increase even if the use of fertilizers is doubled.

X (Fertilizers)	10	20	30	40	50
Y (Production of Grains)	7	12	19	25	35



In the above schedule, there is no specific relationship between the variables. Even though both change in the same direction i.e. both are increasing, they change in different proportions. The ratio of change of variables X and Y in the above schedule is not the same, thus there is a non-linear relationship between the variables

(A) On the basis of number of variables:

1. Simple Correlation:
Simple correlation implies the study between the two variables only. ***For example,*** the relationship between price and demand, and the relationship between price and money supply.

2. Partial Correlation:
Partial correlation implies the study between the two variables keeping other variables constant. ***For example,*** the production of wheat depends upon various factors like rainfall, quality of manure, seeds, etc. But, if one studies the relationship between wheat and the quality of seeds, keeping rainfall and manure constant, then it is a partial correlation.

3. Multiple Correlation:
Multiple correlation implies the study between three or more three variables simultaneously. The entire set of independent and dependent variables is studied simultaneously. ***For example,*** the relationship between wheat output with the quality of seeds and rainfall

Question 4. What is the degree of correlation?

Answer: Degree of Correlation

The degree of correlation is measured through the coefficient of correlation. The degree of correlation for the given variables can be expressed in the following ways:

1. Perfect Correlation:

If the relationship between the two variables is in such a way that it varies in equal proportion (increase or decrease) it is said to be perfectly correlated. This can be of two types:

- **Positive Correlation:** When the proportional change in two variables is in the same direction, it is said to be positively correlated. In this case, the Coefficient of Correlation is shown as **+1**.
- **Negative Correlation:** When the proportional change in two variables is in the opposite direction, it is said to be negatively correlated. In this case, the Coefficient of Correlation is shown as **-1**.

2. Zero Correlation:

If there is no relation between two series or variables, it is said to have zero or no correlation. It means that if one variable changes and it does not have any impact on the other variable, then there is a lack of correlation between them. In such cases, the Coefficient of Correlation will be **0.**

3. Limited Degree of Correlation:

There is a situation with a limited degree of correlation between perfect and absence of correlation. In real life, it was found that there is a limited degree of correlation.

- The coefficient of correlation, in this case, lies between +1 and -1.
- Correlation is limited negative when there are unequal changes in the opposite direction.
- Correlation is limited and positive when there are unequal changes in the same direction.
- The degree of correlation can be **low** (when the coefficient of correlation lies between 0 and 0.25), **moderate** (when the coefficient of correlation lies between 0.25 and 0.75), or **high** (when the coefficient of correlation lies between 0.75 and 1).

Within these limits, the value of correlation can be interpreted as:

Degree of Correlation

Degree of Correlation	Positive Correlation	Negative Correlation
Perfect Correlation	+1	-1
Very High Degree of Correlation	+0.9	-0.9
Fairly High Degree of Correlation	Between +0.75 and +0.9	Between -0.75 and -0.9
Moderate Degree of Correlation	Between +0.25 and +0.75	Between -0.25 and -0.75
Low Degree of Correlation	Between 0 and+0.25.	Between 0 and -0.25.
Zero/No Correlation (uncorrelated)	0	0

Question 5. What is the Karl Pearson's coefficient of correlation?

Answer: The first person to give a mathematical formula for the measurement of the degree of relationship between two variables in **1890 was Karl Pearson**. Karl Pearson's Coefficient of Correlation is also known as **Product Moment Correlation** or **Simple Correlation Coefficient.** This method of measuring the coefficient of correlation is the most popular and is widely used. It is denoted by **'r',** where r is a pure number which means that r has no unit.

*According to **Karl Pearson, "**Coefficient of Correlation is calculated by dividing the sum of products of deviations from their respective means by their number of pairs and their standard deviations."*

Karl Pearson′s Coefficient of Correlation(r)=Sum of Products of Deviations from their respective means / Number of Pairs×(product of Standard Deviations of both Series)

In other words, The ratio of covariance of x & y to the product of standard deviation of x & standard deviation of y , is known as Karl Pearson's coefficient of correlation.

Karl Pearson's coefficient of correlation = Cov(x,y) / $\sigma_x . \sigma_y$

$$r = \sum (x - \bar{x}) (y - \bar{y}) / \sqrt{[\sum(x - \bar{x})^2.\sum(y - \bar{y})^2]}$$

Q6. what is the coefficient of determination?

Answer. In Statistical Analysis, the coefficient of determination method is used to predict and explain the future outcomes of a model. This method is also known as R squared. This method also acts like a guideline which helps in measuring the model's accuracy.

The **coefficient of determination** or R squared method is the proportion of the variance in the dependent variable that is predicted from the independent variable. It indicates the level of variation in the given data set.

- The coefficient of determination is the square of the correlation(r), thus it ranges from 0 to 1.
- With linear regression, the coefficient of determination is equal to the square of the correlation between the x and y variables.
- If R^2 is equal to 0, then the dependent variable cannot be predicted from the independent variable.
- If R^2 is equal to 1, then the dependent variable can be predicted from the independent variable without any error.
- If R^2 is between 0 and 1, then it indicates the extent that the dependent variable can be predictable. If R^2 of 0.10 means, it is 10 percent of the variance in the y variable is predicted from the x variable. If 0.20 means, 20 percent of the variance in the y variable is predicted from the x variable, and so on.

The value of R^2 shows whether the model would be a good fit for the given data set. In the context of analysis, for any given per cent of the variation, it(good fit) would be different. For instance, in a few fields like rocket science, R^2 is expected to be nearer to 100 %. But R^2 = 0(minimum theoretical value), which might not be true as R^2 is always greater than 0(by Linear Regression)

Q7. explain regression analysis.

Answer: Regression analysis is a statistical method used to examine the relationship between a dependent variable and one or more independent variables. It helps determine how changes in the independent variables are associated with changes in the dependent variable, allowing for prediction and understanding of relationships within data.

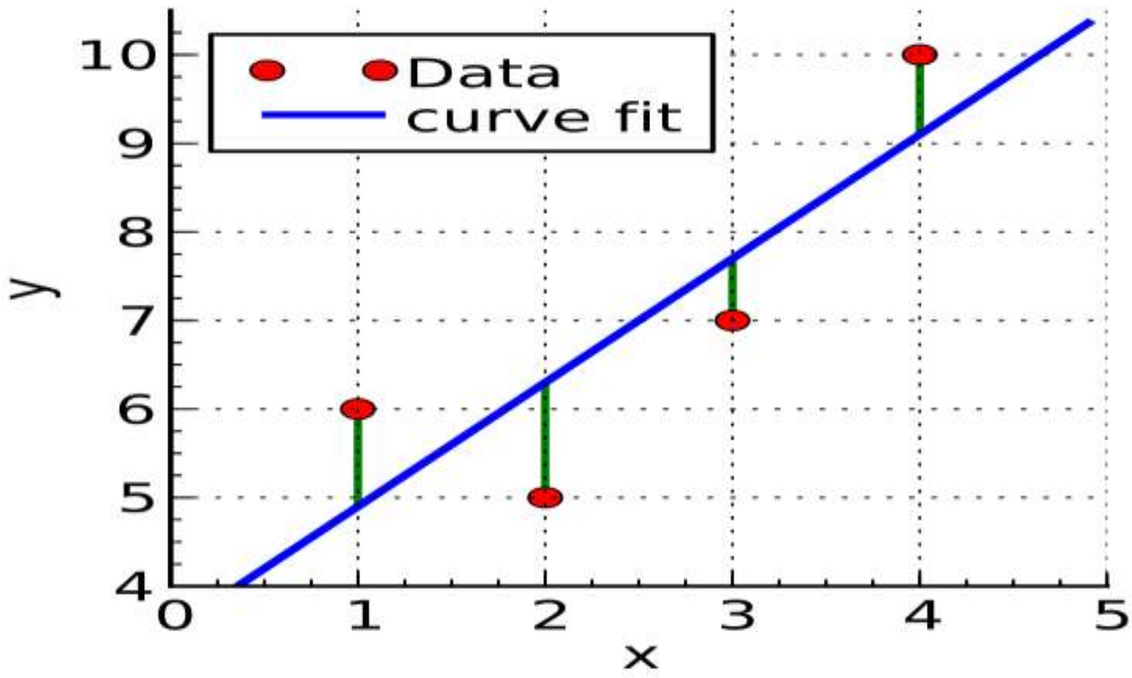
Simple regression: The regression analysis confined to the study of only two variables at a time is termed as the simple regression.

Multiple regression: The regression analysis for studying more than two variables at atime is known as multiple regression.

Linear Regression: Linear Regression is a fundamental statistical method used to define linear relationship between a dependent and one or more independent variables. It helps predict outcomes by fitting a straight line to observed data points, making it easy to interpret and apply.

In statistics, linear regression is a model that estimates the relationship between a scalar response (dependent variable) and one or more explanatory variables (regressor or independent variable). A model with exactly one explanatory variable is a simple linear regression; a model with two or more explanatory variables is a multiple linear regression. This term is distinct from multivariate linear regression, which predicts multiple correlated dependent variables rather than a single dependent variable.

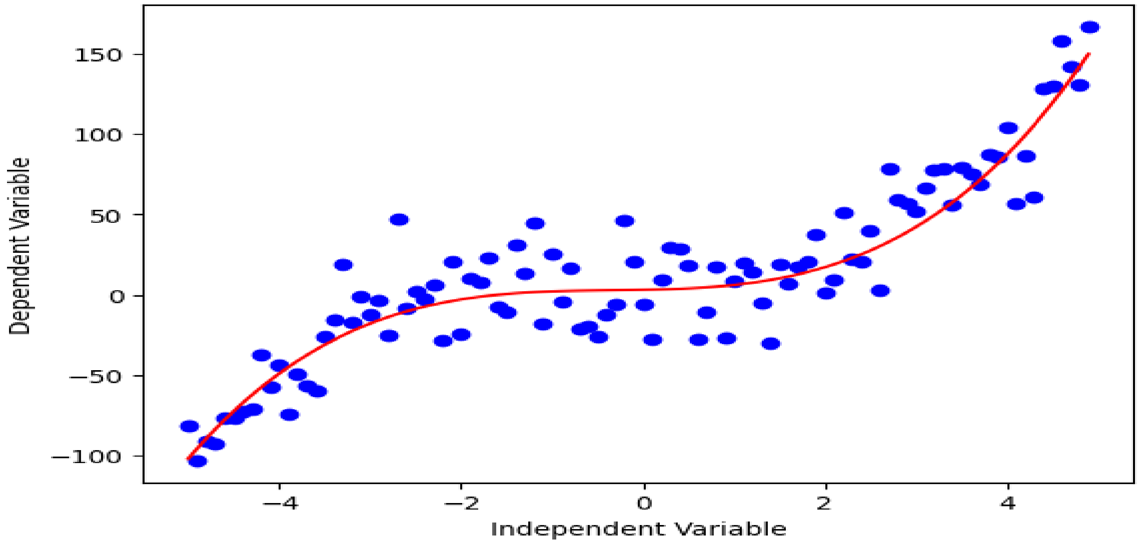
In linear regression, the relationships are modeled using linear predictor functions whose unknown model parameters are estimated from the data. Most commonly, the conditional mean of the response given the values of the explanatory variables (or predictors) is assumed to be an affine function of those values; less commonly, the conditional median or some other quantile is used. Like all forms of regression analysis, linear regression focuses on the conditional probability distribution of the response given the values of the predictors, rather than on the joint probability distribution of all of these variables, which is the domain of multivariate analysis.



Non linear regression: Nonlinear regression refers to a broader category of regression models where the relationship between the dependent variable and the independent variables is not assumed to be linear. If the underlying pattern in the data exhibits a curve, whether it's exponential growth, decay, logarithmic, or any other non-linear form, fitting a nonlinear regression model can provide a more accurate representation of the relationship. This is because in linear regression it is pre-assumed that the data is linear.
A nonlinear regression model can be expressed as:
$Y=f(X,\beta)+\epsilon$
Where,
- $f(X,\beta)$: Regression function
- **X:** This is the vector of independent variables, which are used to predict the dependent variable.
- β :The vector of parameters that the model aims to estimate. These parameters determine the shape and characteristics of the regression function.
- ϵ **:** error term

There are two main types of Non Linear regression in Machine Learning:
(i) Parametric non-linear regression assumes that the relationship between the dependent and independent variables can be modeled using a specific mathematical function. For example, the relationship between the population of a country and time can be modeled using an exponential function. Some common parametric non-linear regression models include: Polynomial regression, Logistic regression, Exponential regression, Power regression etc.
(ii) Non-parametric non-linear regression does not assume that the relationship between the dependent and independent variables can be modeled using a specific mathematical function.

Instead, it uses machine learning algorithms to learn the relationship from the data. Some common non-parametric non-linear regression algorithms include: Kernel smoothing, Local polynomial regression, Nearest neighbor regression etc.



Coefficient of regression

Regression coefficient of y on x: In y = bx + a , the coefficient 'b' which is the slope of line of regression of y on x , is called the coefficient of regression of y on x. It denotes by b_{yx}.

$b_{xy} = \sum(X - \bar{x})(Y - \bar{y}) / \sum(X - \bar{x})^2$

Regression coefficient of y on x: : In x = by + a , the coefficient 'b' which is the slope of line of regression of x on y , is called the coefficient of regression of y on x. It denotes by b_{xy}.

$b_{xy} = \sum(X - \bar{x})(Y - \bar{y}) / \sum(Y - \bar{y})^2$

Q8. Calculate both regression coefficients, coefficient of correlation by regression coefficents & both regression equation for the following data.

X	6	2	10	4	13
Y	9	10	6	8	7

Solution: first of all we calculate mean of x & mean of y.

$\bar{x} = \sum X / n = \{6+2+10+4+13\} / 5 = 35/5 = 7$

$\bar{y} = 40 / 5 = 8$

X	Y	$(X - \bar{x})$	$(Y - \bar{y})$	$(X - \bar{x})(Y - \bar{y})$	$(X - \bar{x})^2$	$(Y - \bar{y})^2$
6	9	6-7 = -1	9-8= 1	-1	1	1
2	10	2- 7= - 5	10-8= 2	-10	25	4
10	6	10-7= 3	6-8= -2	-6	9	4
4	8	4-7= - 3	8-8= 0	0	9	0
8	7	13-7= 8	7-8= -1	-8	64	1
				$\sum(X - \bar{x})(Y - \bar{y}) = -25$	$\sum(X - \bar{x})^2 = 108$	$\sum Y - \bar{y})2 = 10$

(i) coefficient of regression:

$b_{xy} = \sum(X - \bar{x})(Y-\bar{y}) / \sum(Y - \bar{y})^2 = -25 / 10 = -2.5$

$b_{yx} = \sum(X - \bar{x})(Y-\bar{y}) / \sum(X - \bar{x})^2 = -25 / 108 = -0.23$

(ii) coefficient of correlation:

$r^2 = b_{xy} . b_{yx}$

$= (-2.5) . (-0.23)$

$= 0.579$

$r = \sqrt{0.579}$

$r = 0.76$

Since both regression coefficients are negative, so value of are also in negative.

i.e.

$\boxed{r = -0.76}$

(iii) regression equation:

regression line of y on x is

$(y - \bar{y}) = b_{yx} (x - \bar{x})$

$y - 8 = (-0.23)(x - 7)$

$y - 8 = -0.23x + 1.61$

$y = -0.23x + 1.61 + 8$

$\boxed{y = -0.23x + 9.61}$

regression line of y on x is

$(x - \bar{x}) = b_{xy} (y - \bar{y})$

$x - 7 = -2.5 (y - 8)$

$x - 7 = -2.5y + 20.0$

$x = -2.5y + 20 + 7$

$\boxed{x = -2.5y + 27}$

Q9. Calculate Karl Pearson' coefficient of correlation for the following data.

X	**6**	**2**	**10**	**4**	**13**
Y	**9**	**10**	**6**	**8**	**7**

Solution : first of all we calculate mean of x & mean of y.

$\bar{x} = \sum X / n = \{ 6+2+10+4+13\} / 5 = 35/5 = 7$

$\bar{y} = 40 / 5 = 8$

X	Y	$(X - \bar{x})$	$(Y - \bar{y})$	$(X - \bar{x})(Y-\bar{y})$	$(X - \bar{x})^2$	$(Y - \bar{y})^2$
6	9	6-7 = -1	9-8= 1	-1	1	1
2	10	2- 7= - 5	10-8= 2	-10	25	4
10	6	10-7= 3	6-8= -2	-6	9	4
4	8	4-7= - 3	8-8= 0	0	9	0
8	7	13-7= 8	7-8= -1	-8	64	1
				$\sum(X - \bar{x})(Y-\bar{y})=$ -25	$\sum(X - \bar{x})^2 = 108$	$\sum Y - \bar{y})2 = 10$

$$r = \sum (x - \bar{x}) (y - \bar{y}) / \sqrt{} [\sum(x - \bar{x})^2.\sum(y - \bar{y})^2]$$

$$r = -25 / \sqrt{} [108 \times 10]$$

$$r = -25 / \sqrt{} 1080$$

$$r = -25 / 32.86$$

$$\boxed{r = - 0. 76}$$

Which denotes high degree negative correlation.

Q10. Write difference between correlation & regression.

Answer: Difference between correlation and regression

Discerning the distinction between correlation and regression is essential. To better understand how they are used, let's look at some key differences in different aspects.

Relationship

Correlation indicates the possibility of a relationship or association between two variables. It only provides the relationship with strength and direction.

On the other hand, regression is a tool to determine the strength of the correlation between dependent and independent variables. It gives you the ability to quantify this relationship with accuracy. This can give valuable insights into the correlation between them.

Coefficient

In terms of coefficients, correlation and regression differ significantly from one another. Establishing the correlation between two variables is essential in understanding their relationship—how strongly correlated they are. This can be accomplished by examining the signed numerical value of the correlation. The correlation coefficients are between -1.00 and +1.00.

Regression coefficients range from byx >1 to bxy<1, where b is the regression coefficient. Regression coefficients are typically absolute values, whereas correlation coefficients are relative. They must also have the same sign. If byx is positive, bxy must also be positive, for example.

Variables

Correlation and regression are two distinct concepts in which two variables interact. Correlation means that mutual dependence exists between them, while regression shows the impact of the independent variable on the dependent variable.

Cause and effect

It is evident that there is a correlation between the two variables, yet it is not feasible to determine a cause-and-effect relationship. In contrast, regression is based on a cause-and-effect relationship because a change in the values of x (the cause) creates a change in y (effect) values.

Analysis

Correlation analysis is a useful tool for measuring the relationship between two variables; for example, salary levels and employee satisfaction. This helps you see if one is related to the other.

Regression analysis allows you to see how the variables are related. Therefore, you can make predictions and optimise your efforts based on the data results.

Advantages

Correlation analysis is an effective way to summarise the connection between two variables concisely and straightforwardly.

Regression analysis facilitates a detailed examination of the data and includes equations that aid in future prediction and optimisation of the data set.

Usage

Correlation is useful when you need to make a quick judgment based on determining the influence of one variable on another.

Regression becomes necessary when there is a clear correlation between two variables. When a correlation is clear, you only attempt to quantify their connection.

Objectives

Correlation is all about finding the most accurate numerical value to describe the connection between different values, while regression calculates quantitative measures of a random variable with fixed variables. Overall, these two methods help provide useful insights into data analysis.

Similarities between correlation and regression

After going through the main differences, let us now look at the similarities between the two.

- If the correlation between two variables is positive, then the regression slope will be positive.
- If two variables correlate negatively, their regression slope will be downward.
- Usage for both is the same as statistical measurements to fully comprehend the relationship between the variables.

BUDDHA SERIES

(Unit Wise Solved Question & Answers)

Course – BBA 2nd Sem

College – Buddha Institute of Management

(DDU Code-1212)

Department: Business Administration

Subject: Basics of Statistics

Faculty Name: Mr. Abhishek Kumar Gupta

Unit – IV

Q1. What do you mean by time series?

Answer: TIME SERIES

1. A Time Series is defined as a set of observations arranged in Chronological order.
2. An arrangement of Statistical data in accordance with time of occurrence or in a chronological order is called a time series.
3. The numerical or qualitative data which we get at different points of time – the set of observations – is known as time series.
4. Statistically, a Time Series is defined by the values $Y_1, Y_2, \ldots\ldots, Y_n$ of a variable Y at times $t_1, t_2, \ldots\ldots, t_n$. Here Y is a function of time t and Y_t denotes the value of the variable Y at time t.

According to Morris Hamburg- "A time series is a set of observations arranged in Chronological Order"
According to Kenny and Keeping– **"A set of data depending on the time is called time series"**
According to Croxton and Cowden**– "A time series consists of data arranged Chronological"**
According to Patterson – "Time Series consists of Statistical data which are collected, recorded or observed over successive increments"

Examples of Time Series:
i. The Annual Production of steel in India over the last 10 years
ii. The Monthly sales of sugar for the last 6 months
iii. The daily closing price of a share in the Exchange
iv. Hourly temperature recorded by the Meterological office in a city
v. Yearly Price or Quantity Index Number.

Q2. Write use of time series.

Answer: USES OF TIME SERIES
The time series analysis is useful in every sphere. It is very important in economics, business, state administration and planning, science, astronomy, sociology, biology, research work etc., because of the following reasons.
i. It helps in understanding past behavior and it will help in estimating the future behavior
ii. It helps in planning and forecasting. It is very essential in business and economics. With the help of time series we can prepare plans for the future; short range and long range estimates are also possible
iii. Comparison between data of one period with that of another period is possible
iv. We can evaluate the progress in any field of economic and business activity. For example, With the change in price level, we can know the change in the purchasing power of money
v. Seasonal, Cyclical, Secular trend of data is useful not only in economics but also to the businessman.
Thus time series analysis can foretell accurately the course of future event

Q3. Write the name of all components of time series.

Answer: The name of all components of time series are

(i) secular trend or long term movements or long time variation (T)

(ii) Short term variation (a) seasonal variation (S) , (b) Cyclical variation (C)

(iii) Irregular or erratic variation (I)

Q4. Explain Long term fluctuations.

Answer: Long term fluctuations:

- Long term fluctuation is also known as the secular trend. Sometimes it called long time variation.
- A time series data may show upward trend or downward trend for a period of years
- This may be to factors like increase in population, change in technological progress, large scale shift in customer tastes, etc.
- For example, population increases over a period of years
- prices increase over a period of years
- production of goods in the country increases over a period of years.These are examples of upward trend.
- The sales of a commodity may decrease over a period of years because of better products coming to market.
- This is an example of declining trend or downward trend.
- The increase or decrease in ther movements of a time series is called a Secular Trend.

Q4. Explain Seasonal Variations.

Answer: Seasonal Variations

- Seasonal Variations are short-term fluctuation in a time series which occur periodically in a year.
- This continues to repeat year after year.
- The major factors that are responsible for the repetitive pattern of seasonal variations are weather conditions and customs of people.
- Series of monthly and quarterly data are ordinarily used to examine these seasonal variations.
- One can observe that in each year more ice cream are sold during summer season and very less during winter season
- More woolen clothes are sold during winter and very little in summer
- The sales in departmental stores is at peak during Deepavali, Christmas, Pongal months. This reflects the shopping customs of consumers.

Q5. Explain cyclical variation.

Answer: Cyclical variation

- Cyclical Variations are recurrent upward or downward movements in a time series but

the period of cycle is greater than a year.

- These variations are not regular as seasonal variation.
- There are different types of cycles of varying length and size.
- The ups and downs in business activities are the effects of cyclical variation.

・ A business cycle showing these oscillatory movements has to pass through four phases – prosperity, recession, depression and recovery. In a business, these four phases are completed by passing one to another in this order.

Q6. Explain irregular variation.

Answer: Irregular variation

・ Irregular Variations are called as random variation.

・ Random fluctuations in time series that are short in duration, erratic in nature and follow no regularity in the occurrence pattern.

・ These variations are also referred to as residual variations since by definition they represent what is left out in a time series after Trend, Cyclical and Seasonal Variations.

・ Irregular fluctuations result due to the occurrence of unforeseen events like floods, earthquakes, wars etc.

Q7. Explain merits & demerits of the components of time series.

Answer: 1**. Long-Term Movements (Trend)**

Merits:

Forecasting: Identifying trends allows for predicting future values, which is crucial for long-term planning and decision-making in various fields like economics and population growth.

Understanding Underlying Forces: Long-term trends reveal the fundamental drivers of change, such as technological advancements or societal shifts.

Demerits:

Slow to React: Trends develop over extended periods, making them less useful for short-term adjustments or responses to immediate events.

Oversimplification: Focusing solely on the trend can obscure shorter-term fluctuations or cyclical patterns.

2. Cyclical Movements

Merits:

Business Cycle Understanding: Cyclical patterns help analyze economic fluctuations and prepare for expansions and contractions.

Opportunities for Gain: Investors can potentially profit by buying low during recessions and selling high during expansions within a cycle.

Demerits:

Unpredictable Turning Points: Identifying the exact timing of cyclical peaks and troughs can be challenging.

Potential for Misinterpretation: Economic cycles can be complex, and misinterpreting them can lead to poor investment or business decisions.

3. Seasonal Movements

Merits:

Predictable Patterns: Seasonal variations allow businesses to anticipate demand and adjust production, inventory, and staffing accordingly.

Optimized Resource Allocation: Understanding seasonal needs enables efficient allocation of resources, such as labor and materials.

Demerits:

Limited Scope: Seasonal patterns are specific to particular times of the year and don't offer insights into broader trends.

Potential for Over-Reliance: Businesses can become too reliant on predictable seasonal patterns, potentially overlooking unexpected shifts.

4. Irregular Movements

Merits:

Unforeseen Events: Irregular movements reflect unexpected events like natural disasters, political crises, or sudden technological breakthroughs.

Adaptability: Businesses and individuals must be adaptable to respond to these unpredictable changes.

Demerits:

Unpredictable Nature: It is difficult, if not impossible, to anticipate or plan for irregular events.

Potential for Disruption: Irregular movements can cause significant disruptions to established patterns and plans.

Q8. Write the name of four methods of finding trend line.

Answer: The following are some of the methods of finding trend in a time series
i. Free – Hand (Or Graphical) Method
ii. Semi – Average Method
iii. Moving Average Method
iv. Method of Least Squares

SEMI – AVERAGE METHOD:
In this method, the original data is divided into two equal parts and average are
calculated for both the parts. These are averages are called Semi – averages.
For example, we can divide the 10 years 1983 to 1992 into two equal parts; from1983 to 1987 and 1988 to 1992.
If the period is odd number of years, the values of the middle year is omitted say, from 1983 to 1993 we must omit the year 1988.We can draw the line by a straight line by joining the two points of averages. By

extending the line downwards or upwards we can get the intermediate values or we can predict the future.

Merits of Semi – Average Method:

i. It is simple and easier to understand than moving average and least square method
ii. As the line can be extended both ways, we can get the intermediate values and predict the future values.
iii. As it does not depend upon personal judgement, everyone who applies this method will get the same trend line unlike the former method.

Demerits of Semi – Average Method:

i. Under this method, it has an assumption of linear trend whether such a relationship exists or not
ii. It is affected by the limitation of arithmetic mean
iii. This method is not enough for forecasting the future trend or for removing trend from

MOVING – AVERAGE METHOD:

Moving average method is a simple device of reducing fluctuations and obtaining trend values with a fair degree of accuracy. In this method the average value of number of years (or months, weeks or days) is taken as the trend value for the middle point of the period of moving average. The process of averaging smoothes the curve and reduces the fluctuation.
The first thing to be decided in the method is the period of the moving average. What it means is to take a decision about the number of consecutive items whose average would be calculated each time.
Suppose it has been decided that the period of the moving average would be 5 years or months or weeks or days (as the case may be) then the arithmetic average of the first 5 items (number 1, 2, 3, 4 and 5) would be placed against item No. 3 and then the arithmetic average of item Nos, 2,3,4,5 and6 would be placed against item No.4. This process would be repeated till the arithmetic average of the last 5 items has been calculated.

Odd Period of Moving Average:

Steps for calculating odd number of years (3,5,7,9), e.g., calculation of three yearly moving average include the following steps:
1. Compute the value of first three years (1,2,3) and place the three year total against the middle year (i.e., 2nd year)
2. Leave the first year's value and add up the values of the next three years i.e., 2,3,4 and place the three-year total against the middle year i.e., 3rd year
3. This process must be continued until the last year's value is taken for calculating moving average.
4. The three-yearly total must be divided by 3 and placed in the next column. This is the trend value of moving average.

The formula calculating3 yearly moving average is as follows :
(a+b+c)/3, (b+c+d)/3, (c+d+e)/3
The formula for 5 yearly moving average is as follows :
(a+b+c+d+e)/5, (b+c+d+e+f)/5, (c+d+e+f+g)/5

Even Period of Moving Average:

If the period of moving average is 4, 6 or 8, it is an even number. The four-yearly total cannot be place against any year as the median 2.5 is between the second and the third year. So the total should be placed in between the 2nd and 3rd years. We must centre the moving average in order to place the moving average against the year.
The steps are:
i. Compute the values of the first four years and place the total in between the 2nd and the 3rd years
ii. Leave the first year value and compute the value of the next four years and

place the total in between the 3rd and 4th year
iii. This process must be continued until the last year is taken into account
iv. Compute the first two four-year totals and place it against the middle year (i.e., 3rd year)
v. Leave the first four year total and computeriginal data.

Q9. **(a) (For Odd Number of Items)**

Fit a trend line by the method of semi-averages for the given data.

Year	2000	2001	2002	2003	2004	2005	2006
Production	105	115	120	100	110	125	135

Solution:

Since the number of years is odd(seven), we will leave the middle year's production value and obtain the averages of first three years and last three years.

Year	Production	Average
2000	105	
2001	115	$\frac{105+115+120}{3}=113.33$
2002	120	
2003	100 (left out)	
2004	110	
2005	125	$\frac{110+125+135}{3}=123.33$
2006	135	

Table 9.1

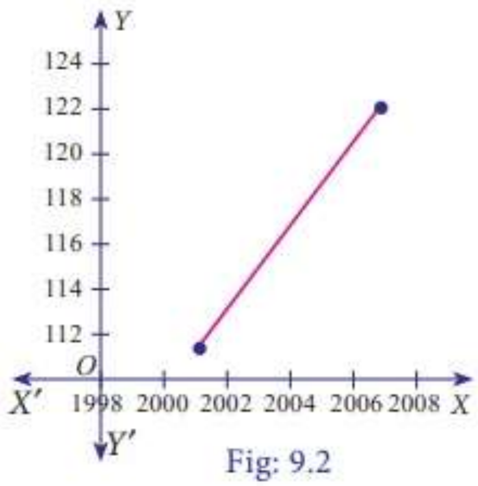


Fig: 9.2

. **(b) (For even Number of Items)**

Fit a trend line by the method of semi-averages for the given data.

Year	1990	1991	1992	1993	1994	1995	1996	1997
Sales	15	11	20	10	15	25	35	30

Solution:

Since the number of years is even(eight), we can equally divide the given data it two equal parts and obtain the averages of first four years and last four years.

Year	Production	Average
1990	15	$\frac{15+11+20+10}{4}=14$
1991	11	
1992	20	
1993	10	
1994	15	$\frac{15+25+35+30}{4}=26.25$
1995	25	
1996	35	
1997	30	

Table 9.2

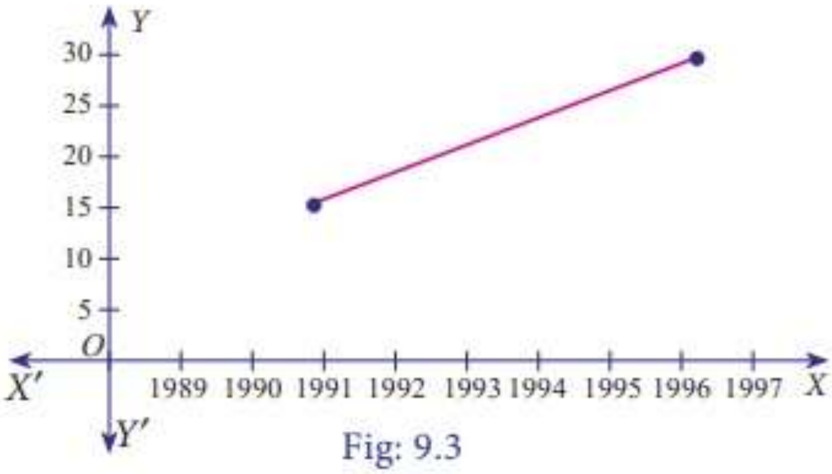


Fig: 9.3

Q10(a). **Calculate three-yearly moving averages of number of students studying in a higher secondary school in a particular village from the following data.**

Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004
Number of students	332	317	357	392	402	405	410	427	435	438

Solution:

Computation of three- yearly moving averages.

Year	Number of students	3- yearly moving Total	3- yearly moving Averages
1995	332	---	---
1996	317	1006	335.33
1997	357	1066	355.33
1998	392	1151	383.67
1999	402	1199	399.67
2000	405	1217	405.67
2001	410	1242	414.00
2002	427	1272	424.00
2003	435	1300	433.33
2004	438	---	---

Table 9.3

Q10(b). Calculate four-yearly moving averages of number of students studying in a higher secondary school in a particular city from the following data.

Year	2001	2002	2003	2004	2005	2006	2007	2008
Sales	124	120	135	140	145	158	162	170

Solution:

Computation of four- yearly moving averages.

Year	Sales	4-yearly centered moving total	4-yearly moving Average	4-yearly centered moving Average
2001	124	--	--	--
2002	120	--	--	--
		519	129.75	
2003	135	--		132.37
		540	135.00	
2004	140	--		139.75
		578	144.50	
2005	145	--		147.87
		605	151.25	
2006	158	--		155.00
		635	158.75	
2008	162	--		162.50
		665	166.25	
2007	170	--	--	-
2008	175	--	--	-

Table 9.4

Note: The calculated 4-yearly centered moving average belongs to the particular year present in that row eg; 132.37 belongs to the year 2003.

BUDDHA SERIES

(Unit Wise Solved Question & Answers)

Course – BBA 2nd Sem

College – Buddha Institute of Management

(BIM Code-1212)

Department: Business Administration

Subject: Basics of Statistics

Faculty Name: Mr. Abhishek Kumar Gupta

Unit – V

Q1. What do you mean by index numbers?

Answer: Meaning of Index Numbers: The value of money does not remain constant over time. It rises or falls and is inversely related to the changes in the price level. A rise in the price level means a fall in the value of money and a fall in the price level means a rise in the value of money. Thus, changes in the value of money are reflected by the changes in the general level of prices over a period of time.

The index number was first constructed by an Indian Statistician, Carli in 1764. It was used for the first time to compare the prices of the year 1750 with that of the year 1500. An index number is a statistical tool for measuring changes in the magnitude of a group of related variables.

Index number is a technique of measuring changes in a variable or group of variables with respect to time, geographical location or other characteristics.

An index number is a statistical device for measuring changes in the magnitude of a group of related variables. It represents the general trend of diverging ratios, from which it is calculated. It is a measure of the average change in a group of related variables over two different situations. The comparison may be between like categories such as persons, schools, hospitals etc. An index number also measures changes in the value of the variables such as prices of specified list of commodities, volume of production in different sectors of an industry, production of various agricultural crops, cost of living etc.

In economics, statistics, and finance, an **index** is a number that measures how a group of related data points—like prices, company performance, productivity, or employment—changes over time to track different aspects of economic health from various sources.

The **index number** helps in studying changes in consumption, production, imports, exports, cost of living, crimes, accidents, national income, business failures, and other phenomena.

"Index Number shows by its variation the changes in a magnitude which is not susceptible either of accurate measurement in itself or of direct valuation in practice."**- Edgeworth**

"Index Numbers are devices for measuring differences in the magnitude of a group of related variables."**- Croxton and Cowden**

"An index number is a statistical measure designed to show changes in variable or a group of related variables with respect to time, geographical location or other characteristics." **- Spiegel**

An index number is used to display changes in a variable or group of related variables through time, space, or other factors. Comparisons can be made between elements like people, hospitals, schools, etc. It tracks changes in the value of variables such as the cost of living, the volume of production in various industries, the prices of a list of defined commodities, and so on. Also, as the index numbers are used to feel the pulse of the economy, they are also known as the **Barometer of economic activity.**

Q2. What are issues (problems) in the construction of index numbers?
Answer: There are three major issues which may be faced in the construction of index numbers. They are: 1) Collection of Data; 2) Selection of Base Year and 3) Selection of Appropriate Index. Let us discuss them in detail:
1) Collection of Data

Data collection through a sample method is one of the issues in the construction of index numbers. The data has to be as reliable, adequate, accurate, comparable, and representative, as possible. Here a large number of questions need to be answered. The answers ultimately depend on the purpose and individual judgement. For example, one needs to decide the following

i) **Identification of Commodities to be Included:** How many and which category of commodities to include? A large number of items may be present. It is not possible to include all of them, only those items deserve to be included in the construction of an index number as would make it more representative. For example, if we are required to construct indices for shares on the Bombay Stock Exchange, there are several shares listed and traded, it is not possible to include all of them. Therefore, it has to be decided which sample number of shares (may be 30 or 40) should represent the general movement of share prices of the Bombay Stock Exchange. Therefore, it is worthwhile to note that the selection of items must be deliberate and in keeping with the relevance and significance of each individual item to the purpose for which the index is constructed.

ii) **Sources of Data:** From where to collect data? It is an important and difficult issue. The source depends on the information requirement. For example, one may need to collect prices and quantities consumed related to certain commodities for a consumer price index. However, there may be a large number of retailers and wholesalers, selling the commodities, and quoting different prices. To get the details, only a few representative shops (which represent the typical purchasing points of the people under question) need to be selected. Thus, based on a representative sample survey, sources should be from where accurate, adequate, and timely data can be available.

iii) **Timings of Data Collection:** It is also equally important to collect the data at an appropriate time. Referring to the example of consumer price index, prices are likely to vary on different days of the month. For certain commodities prices may vary at different times of the same day. Take an example, vegetable prices are usually high in the morning when fresh vegetables arrive and are low in the late evening when sellers are closing for the day and wish to clear the perishable stock. For each commodity, individual judgement needs to be exercised to represent reality and to serve the purpose for which an index is to be used.

2) Selection of Base Year

A base period is the reference period for comparing and analysing the changes in prices or quantities in a given period. For many index number series, value of a particular time period, usually a year, is taken as reference period against which all subsequent index numbers in the series are calculated and compared. In some other cases, especially when cost of living needs to be compared across the cities, the value of cost of living prevailing in a selected city is taken as a base against which cost of living in other cities is compared.

In yet other cases, we may be required to compare one index number series against another series. In such a context, a 'base' common to all series is more appropriate.

In the light of the above considerations, therefore, the period/year selected as base period/year must be a 'normal' period. **Normal period** is a period with price or quantity figures neither too low, nor too high. It should not have been affected by abnormal occurances, such as floods, (if interested in agricultural production), wars, sudden recession etc. What is normal should also be decided keeping in view the purpose of constructing an index number, and the specific situation.

3) Selection of an Appropriate Index

Different methods of indices give different results, when applied to the same data. Utmost care must be taken in selection of a formula which is the most suitable for the purpose. Whether to use an unweighted or weighted index is a difficult question to answer. It depends on the purpose for which the index number is required to be used. For example, if we are interested in an index for the purpose of negotiating wages or compensating for price rise, only a weighted index would be worthwhile to use.

Q3. Write the name of all types of index numbers.

Answer: There are three principal types of indices

(i) Price indices

(ii)Quantity indices

(iii) Value indices

Q4. Explain price index mubers.

Answer: Price indices: This type of indices is the most frequently used. Price indices consider prices of a commodity or a group of commodities and compare changes of prices from one period to another period and also compare the difference in price from one place to another. For example, the familiar Consumer Price Index measuring overall price changes of consumer commodities and services is used to define the cost of living.

Weighted Aggregative Method Of Price Index: When the relative importance of items is considered, an index number becomes a weighted index. This index is calculated after assigning weights to the commodities on the basis of their importance. Weights assigned are then multiplied to their base and current period prices to get weighted prices. The sum of weighted price to the sum of weighted prices of the base year multiplied by 100 is called a weighted **aggregative** price index.

Weighted aggregative price index can be calculated using different formulas for the same problem. Some of the important methods are

Laspeyre's price index: The Laspeyre's price index is an index formula used in price statistics for measuring the price development of the basket of goods and services consumed in the base period. The question it answers is how much a basket that consumers bought in the base period would cost in the current period. It is defined as a fixed-weight, or fixed-basket, index that uses the basket of goods and services and their weights from the base period. It is also known as a "**base-weighted index**".

'' The method of calculating Weighted Index Numbers under which the base year quantities are used as weights of different items is known as **Laspeyre's Method.''** The formula for Laspeyre's Price Index is:

Laspeyre's Price Index (P_{01}) = $\sum p_1q_0 / \sum p_0q_0 \times 100$

Paasche Price Index: The Paasche Price Index is a consumer price index used to measure the change in the price and quantity of a basket of goods and services relative to a base year price and observation year quantity. Developed by German economist Hermann Paasche, the Paasche Price Index is commonly referred to as the "current weighted index."

The Paasche Price Index is a price index used to measure the general price level and cost of living in the economy and calculate inflation. The index commonly uses a base year of 100, with periods of higher price levels shown by an index greater than 100 and periods of lower price levels by indexes lower than 100.

The Paasche Price Index is commonly confused with the Laspeyres Price Index. The main differentiator between the Paasche Index and the Laspeyres Price Index is that the former uses current-period quantity weightings while the latter uses base-period quantity weightings.

Paache's Price Index (P_{01}) = $\sum p_1q_1 / \sum p_0q_1 \times 100$

Fisher Price Index: The Fisher Price Index, also called the Fisher's Ideal Price Index. It is a consumer price index (CPI) used to measure the price level of goods and services over a given period. The Fisher Price Index is a geometric average of the Laspeyres Price Index and the Paasche Price Index. It is deemed the "ideal" price index as it corrects the positive price bias in the Laspeyres Price Index and the negative price bias in the Paasche Price Index.

Fisher's Price Index (P_{01}) = √ (Laspeyre's Price Index) × (Paache's Price Index)

or

Fisher's Price Index (P_{01}) = $\sqrt{[(\sum p_1q_0 / \sum p_0q_0) \times (\sum p_1q_1 / \sum p_0q_1)]} \times 100$

Q5. Explain Quantity index numbers.

Answer: Quantity indices: Quantity indices measure changes in the physical amounts of goods or services produced, consumed, or traded over time. They are distinct from price indices (which track price changes) and value indices (which combine price and quantity changes).

A **volume index** or **quantity index** is a numerical time series measure designed to help compare how the production of some class of goods and/or services, taken as a whole, differs between time periods or geographical locations.

A quantity index number is a measure of changes in quantity between time periods. These index numbers are often used to measure things such as employment, production, or construction. An example would be an index that measures quantity changes in industrial production.

Laspeyre's quantity index: The Laspeyres quantity index measures changes in the quantity of goods or services between two periods using the prices of the base period as weights.

Laspeyre's quantity index (Q_{01}) = $\sum q_1p_0 / \sum q_0p_0 \times 100$

Paasche's quantity index: The Paasche quantity index is a statistical measure that compares the quantity of goods and services produced or consumed in a current period to the quantity in a base period, using the current period's prices as weights.

Paasche's quantity index (Q_{01}) = $\sum q_1p_1 / \sum q_0p_1 \times 100$

Fisher's quantity Index: The geometric mean of Laspeyre's quantity index & Paasche's quantity index is known as the Fisher's quantity Index.

Fisher's quantity Index (Q_{01}) = √ (Laspeyre's quantity Index) × (Paache's quantity Index)

or

Fisher's Quantity Index (Q_{01}) = $\sqrt{(\sum q_1p_0 / \sum q_0p_0) \times (\sum q_1p_1 / \sum q_0p_1)]} \times 100$

Q6. What is the value index number?

Answer: Value index numbers: A value index number measures changes in the aggregate value of a variable or group of variables over time. Value index numbers are used to track changes in things such as trade, inventories, and sales.

A value index number is a statistical measure that shows the relative change in the total value of a group of items (like goods or services) between two periods, typically a base period and a current period. It's calculated by comparing the total value (price multiplied by quantity) in the current period to the total value in the base period.

The value index number reflects changes in both price and quantity of the items being measured. It indicates how much the overall value of those items has increased or decreased between the two periods.

It's not just about price changes (like a price index) or quantity changes (like a quantity index), but the combined effect of both.

The ratio of $\sum p_1q_1$ & $\sum p_0q_0$ with multiplication 100, is known as the value index number.

i.e.

Value index number (V_{01}) = $(\sum p_1q_1 / \sum p_0q_0) \times 100$

Q7. Explain tests of adequacy of index numbers.

Answer: Time Reversal Test: It is an important test for testing the consistency of a good index number. This test maintains time consistency by working both forward and backward with respect to time (here time refers to base year and current year).

The price index number of current year with respect to base year is the reciprocal of the price index number of base year with respect to current year.

i.e. $P_{01} = 1 / P_{01}$

$P_{01} \times P_{01} = 1$

There are five methods which do satisfy the test:

(1) The Fisher's ideal formula,

(2) Simple geometric mean of price relatives,

(3) Aggregates with fixed weights,

(4) The weighted geometric mean of price relatives if we use fixed weights, and

(5) Marshall-Edgeworth method.

We show Fisher's ideal index satisfies the test

$$P_{01}=\sqrt{\frac{\sum p_1q_0}{\sum p_0q_0}\times\frac{\sum p_1q_1}{\sum p_0q_1}}$$

Changing time, i.e., 0 to 1 and 1 to 0.

$$P_{10}=\sqrt{\frac{\sum p_0q_1}{\sum p_1q_1}\times\frac{\sum p_0q_0}{\sum p_1q_0}}$$

$$P_{01}\times P_{10}=\sqrt{\frac{\sum p_1q_0}{\sum p_0q_0}\times\frac{\sum p_1q_1}{\sum p_0q_1}\times\frac{\sum p_0q_1}{\sum p_1q_1}\times\frac{\sum p_0q_0}{\sum p_1q_0}} = \sqrt{1} =1.$$

Since P01 X P10 = 1, thus Fisher's ideal index satisfies the test.

Factor Reversal Test: This is another test for testing the consistency of a good index number. The product of price index number and quantity index number from the base year to the current year should be equal to the true value ratio. That is, the ratio between the total value of current period and total value of the base period is known as true value ratio. Factor Reversal Test is given by,

$$P_{01}\times Q_{01}=\frac{\sum p_1q_1}{\sum p_0q_0}$$

where $$P_{01}=\sqrt{\frac{\sum p_1q_0\times\sum p_1q_1}{\sum p_0q_0\times\sum p_0q_1}}$$

Now interchanging P by Q, we get

$$Q_{01}=\sqrt{\frac{\sum q_1p_0\times\sum q_1p_1}{\sum q_0p_0\times\sum q_0p_1}}$$

where P_{01} is the relative change in price.

Q_{01} is the relative change in quantity.

Circular Test: Another test of the adequacy of index number formula is what is known as 'circular test'. If in the use of index numbers interest attaches not merely to a comparison of two years, but to the measurement of price changes over a period of years. It is frequently desirable to shift the base. Clearly, the desirability of this property is that it enables us to adjust the index values from period to period without referring each time to the original base. A test of this shift ability of base. A test of this shift ability of base is called to the circular test.

This test is just an extension of the time reversal test. The test requires that if an index is constructed for the year a on base year b, and for the year b on base year c, we ought to get the same result as if we calculated direct an index for a on base year c without going through b as an intermediary.

Symbolically, if there are three indices P01, P12 and P20 the Circular test will be satisfied if

$P_{01} \text{ X } P_{12} \text{ X } P_{20} = 1$

The Laspeyre's index does not satisfy the test as can be seen from the following:
If the three years are 0, 1, 2, the index by Laspeyre's method will be

$$\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_2 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_2}{\sum p_2 q_2}$$

The product of all these is not equal to 1. Hence the test is not satisfied.

Q9. Calculate Fisher's price index number and show that it satisfies both Time Reversal Test and Factor Reversal Test for data given below.

Commodities	Price		Quantity	
	2003	2009	2003	2009
Rice	10	13	4	6
Wheat	15	18	7	8
Rent	25	29	5	9
Fuel	11	14	8	10
Miscellaneous	14	17	6	7

Solution:

Commodities	Price		Quantity		p_0q_0	p_0q_1	p_1q_0	p_1q_1
	2003 (p_0)	2009 (p_1)	2003 (q_0)	2009 (q_1)				
Rice	10	13	4	6	40	60	52	78
Wheat	15	18	7	8	105	120	126	144
Rent	25	29	5	9	125	225	145	261
Fuel	11	14	8	10	88	110	140	140
Miscellaneous	14	17	6	7	84	98	102	119
				Total	442	613	565	742

Table 9.11

Fisher's price index number

$$P_{01}^{F} = \left(\sqrt{\frac{\sum p_1q_0 \times \sum p_1q_1}{\sum p_0q_0 \times \sum p_0q_1}}\right) \times 100 = \left(\sqrt{\frac{565 \times 742}{442 \times 613}}\right) \times 100 = 124.3898$$

Time Reversal Test: $P_{01} \times P_{10} = 1$

$$P_{01} \times P_{10} = \sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum p_0q_1 \times \sum p_0q_0}{\sum p_0q_0 \times \sum p_0q_1 \times \sum p_1q_1 \times \sum p_1q_0}\right)}$$

$$P_{01} \times P_{10} = \sqrt{\left(\frac{565 \times 742 \times 613 \times 442}{442 \times 613 \times 742 \times 565}\right)}$$

$$P_{01} \times P_{10} = 1$$

Factor Reversal Test

$$P_{01} \times Q_{01} = \frac{\sum p_1q_1}{\sum p_0q_0}$$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum q_1p_0 \times \sum q_1p_1}{\sum p_0q_0 \times \sum p_0q_1 \times \sum q_0p_0 \times \sum q_0p_1}\right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{565 \times 742 \times 613 \times 742}{442 \times 613 \times 442 \times 565}\right)}$$

$$P_{01} \times Q_{01} = \sqrt{\left(\frac{742 \times 742}{442 \times 442}\right)} = \frac{742}{442} \quad \Rightarrow P_{01} \times Q_{01} = \frac{\sum p_1q_1}{\sum p_0q_0}$$

Q10.Calculate Fisher's price index number and show that it satisfies both Time Reversal Test and Factor Reversal Test for data given below.

Commodities	Base Year		Current Year	
	Price	Quantity	Price	Quantity
Rice	10	5	11	6
Wheat	12	6	13	4
Rent	14	8	15	7
Fuel	16	9	17	8
Transport	18	7	19	5
Miscellaneous	20	4	21	3

Solution:

Commodities	Base Year		Current Year		p_0q_0	p_0q_1	p_1q_0	p_1q_1
	Price (p_0)	Quantity (q_0)	Price (p_1)	Quantity (q_1)				
Rice	10	5	11	6	50	60	55	66
Wheat	12	6	13	4	72	48	78	52
Rent	14	8	15	7	112	98	120	105
Fuel	16	9	17	8	144	128	153	136
Transport	18	7	19	5	126	90	133	95
Miscellaneous	20	4	21	3	80	60	84	63
				Total	584	484	623	517

Table 9.12

Fisher's price index number

$$P_{01}^{F}=\left(\sqrt{\frac{\sum p_1q_0 \times \sum p_1q_1}{\sum p_0q_0 \times \sum p_0q_1}}\right)\times 100=\left(\sqrt{\frac{623\times 517}{584\times 484}}\right)\times 100=106.74$$

Time Reversal Test: $P_{01}\times P_{10}=1$

$$P_{01}\times P_{10}=\sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum p_0q_1 \times \sum p_0q_0}{\sum p_0q_0 \times \sum p_0q_1 \times \sum p_1q_1 \times \sum p_1q_0}\right)}$$

$$P_{01}\times P_{10}=\sqrt{\left(\frac{623\times 517\times 484\times 584}{584\times 484\times 517\times 623}\right)}$$

$$P_{01}\times P_{10}=1$$

Factor Reversal Test

$$P_{01}\times Q_{01}=\frac{\sum p_1q_1}{\sum p_0q_0}$$

$$P_{01}\times Q_{01}=\sqrt{\left(\frac{\sum p_1q_0 \times \sum p_1q_1 \times \sum q_1p_0 \times \sum q_1p_1}{\sum p_0q_0 \times \sum p_0q_1 \times \sum q_0p_0 \times \sum q_0p_1}\right)}$$

$$P_{01}\times Q_{01}=\sqrt{\left(\frac{623\times 517\times 484\times 517}{584\times 484\times 584\times 623}\right)}$$

$$P_{01}\times Q_{01}=\sqrt{\left(\frac{517\times 517}{585\times 584}\right)}=\frac{517}{584}$$

$$P_{01}\times Q_{01}=\frac{\sum p_1q_1}{\sum p_0q_0}$$

Q11. What are limitations of index numbers?

Answer: LIMITATIONS OF INDEX NUMBERS:

International Comparison is not possible: International comparisons are not possible on the basis of Index Numbers as different countries have different basis of index numbers. Due to differences in the unit of currency and also difference in the composition of production and consumption across the different countries, index numbers fail to facilitate international comparisons.

Limited use of Index Numbers: Index Numbers have limited uses because they are constructed with certain specific purpose and they can not give information about other facts. For example, Index Number prepared for the study of standard of living of industrial workers can not give knowledge about the standard of living of agricultural labourers.

Approximate Indicators: Index Numbers point out the general trend. They simply present approximate view of the problem under study, so whatever is pointed by Index Number, should not be taken as granted. It should be viewed as an approximate.

Difference of Time: With the passage of time, it is difficult to make comparisons of Index Numbers. With the changing times, consumers habits, tastes, etc. , also undergo a change . Consequently, Index Numbers constructed on the basis of old consumption pattern can not be compared with the Index Numbers constructed on the basis of new consumption pattern.

Based on Samples: Index numbers are generally based on samples, because it is not possible to take into account each and every item in the construction of the index numbers. Index number generally do not take into account changes in the quality of products. Index Numbers suffer from the limitations of the methods used for their construction. Each methods has its own merits and limitations. No particular method is suitable for all circumstances.

Manipulation possible: Index Numbers can be manipulated in such a manner as to draw desired conclusions. The selection of a base year, commodities, prices, etc., may help a person to prove his viewpoint.

Not completely True: Index numbers are not fully true. For example, one can only make an estimate of change in the value of money with the help of index numbers. The Index Numbers simply indicate arithmetical tendency of the temporal changes in the variable.